

FIT3155: Week 3 Lab Sheet

(Questions are based on Week 2 topic, the Boyer-Moore algorithm)

Objectives: Develop a clear understanding and working implementation of the Boyer-Moore algorithm.

Conceptual exercises

- ☐ 1. Consider the following pattern string drawn from the alphabet $\Sigma = \{x, y, z\}$:

1 2 3 4 5 6 7 8 9 10 11
y z z y z x y z z y z

Work out (on paper) (i) standard bad-character shift array and (ii) the extended bad-character shift table for the above pattern.

- ☐ 2. Determine the worst-case time and space complexities to **construct**:
- the standard bad-character shift array, and
 - the extended bad-character shift table.

Separately, after construction, what is the worst-case time to **retrieve** any shift value from each of the above?

- ☐ 3. For a pattern of length m , can a data structure supporting the **extended** bad-character shift function (beyond the one that was realized as an $m \times |\Sigma|$ table as we saw in the lecture) be constructed in $O(m + |\Sigma|)$ worst-case time and space?

If Yes: How? And what is the worst-case access time to **retrieve** shift values from this construction?

If No: Why not?

4. Consider the following pattern string drawn from the alphabet $\Sigma = \{a, b, c\}$:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
a b a a a b a c b a a b a a a b

Construct for the above pattern by hand

- ☐ (a) its good-suffix shift array, and
- ☐ (b) its matched-prefix shift array.

- 5. Consider the following **periodic** text and pattern over the alphabet $\Sigma = \{a, b, c, d, e\}$:

```

      1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
txt = a b c d e a b c d e a b c d e a b c d e a b c d e a b c d e a b c d e a b c d e
      1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
pat = a b c d e a b c d e a b c d e

```

Using the mechanics of Boyer-Moore algorithm applicable for this scenario, work out how many explicit character comparisons are performed to find all occurrences of the pattern in the text?

- 6. Consider some iteration of the Boyer-Moore algorithm involving the pattern $\text{pat}[1 \dots m]$ and the text $\text{txt}[1 \dots n]$. Suppose that in this iteration the region $\text{txt}[j \dots j + m - 1]$ of the text is being compared with pat .

Prove that if a right-to-left scan results in a mismatch at position $k \leq m$ and the value in the good-suffix array satisfies $\text{gs}[k + 1] > 0$ ^{*} (using 1-based indexing note), then the shift by $m - \text{gs}[k + 1]$ using the good suffix rule is **safe**, i.e., such a shift can never move pat past a valid occurrence in txt .

- 7. Consider some iteration of the Boyer-Moore algorithm involving the pattern $\text{pat}[1 \dots m]$ and the text $\text{txt}[1 \dots n]$. Suppose that in this iteration the region $\text{txt}[j \dots j + m - 1]$ of the text is being compared with pat .

Prove that if a right-to-left scan results in a mismatch at position $k \leq m$ and the value in the good-suffix array satisfies $\text{gs}[k + 1] = 0$ ^{ps} (using 1-based indexing note), then the shift by $m - \text{mp}[k]$ using the matched prefix rule is **safe**, i.e., such a shift can never move pat past a valid occurrence in txt .

- 8. Consider some iteration of the Boyer-Moore algorithm involving the pattern $\text{pat}[1 \dots m]$ and the text $\text{txt}[1 \dots n]$. Suppose that in this iteration the region $\text{txt}[j \dots j + m - 1]$ of the text is being compared with pat .

Prove that if a right-to-left scan results in $\text{pat}[1 \dots m] = \text{txt}[j \dots j + m - 1]$, then the shift by $m - \text{mp}[2]$ (in 1-based indexing, note) using the special application of matched prefix rule is **safe**, i.e., such a shift can never move pat past another such occurrence in txt .

*It is safe to assume that when $k = m$ the implied good suffix is a **null** string, and there is an additional cell $\text{gs}[m + 1]$ preprocessed as part of gs array storing a valid value for shifting safely, as discussed in the lecture.

Implementation exercises

(A message of caution: ‘Vibe coding’ with GenAI will hinder your learning of algorithms and data structures. The degree (and this unit) you have enrolled into is a specialist degree (unit). By definition it expects one to demonstrate specialist understanding of topics. This understanding comes from making a concerted effort, via thinking, implementing, debugging etc.)

- ☒ 1. In preparation for implementing the Boyer-Moore algorithm for exact pattern matching, explore what optimisations your implementation could benefit from and how they might improve the running time, on average and in the worst case.
- ☒ 2. Implement Boyer-Moore’s algorithm for pattern matching. Your script accepts files containing the text and pattern as inputs, and outputs the positions of all instances of the pattern observed in that text.

```
--o0o--  
END  
--o0o--
```